F18CE2 MV Calculus and Real Analysis B: Tutorial 2

1 Suppose that (a_n) and (b_n) are sequences with $|a_n| \leqslant K|b_n|$ for all n, where $K > 0$, and $\lim_{n\to\infty} b_n = 0$.

Prove directly from the definition of limit that $\lim_{n\to\infty} a_n = 0$.

2 Suppose that (a_n) and (b_n) are convergent sequences with $\lim_{n\to\infty} a_n = L$ and $\lim_{n\to\infty} b_n = M$.
Prove directly from the definition of limit that $\lim_{n\to\infty} (a_n + b_n) = L + M$.

3 Suppose that $\lim_{n\to\infty} a_n = L$. Prove directly from the definition of limit that there exists an integer N such that $|a_n| \leqslant 1 + |L|$ for all $n \geqslant N$.

4 Suppose that $\lim_{n\to\infty} a_n = L > 0$. Prove directly from the definition of limit that there exists an integer N such that $a_n > \dfrac{L}{2}$ for all $n \geqslant N$.

If $a_n > 0$ for all n, prove directly from the definition of limit that $\lim_{n\to\infty} \dfrac{1}{a_n} = \dfrac{1}{L}$.

5 Let A be a non-empty set of real numbers. Define what is meant by A is bounded above and what is the supremum of A.

6 True or false?

(a) Every non-empty set of real numbers has a supremum.
(b) If M is an upper bound for a set A and $L > M$, then L is also an upper bound for A.

7 Let A and B be bounded sets of real numbers with $\sup A = \sup B$ and $\inf A = \inf B$. Does this imply that $A = B$?

8 Let $A = \{x \in \mathbb{R} : x < 2\}$. Find the supremum of A and prove that your answer is correct.

9 Let A be a non-empty set of real numbers that is bounded above.

Let B be the set consisting of numbers of the form $b = 3a + 1$, for $a \in A$.

Show that B is bounded above and that $\sup B = 3 \sup A + 1$.

10 Let A be a non-empty set of real numbers that is bounded above, and such that $a > 0$ for all $a \in A$.

Let B be the set consisting of numbers of the form $b = 1/(1 + a)$, for $a \in A$.

Show that B is bounded below and that

$$\inf B = \frac{1}{1 + \sup A}.$$